# Will AI Agents really Automate Jobs at Scale?

By Jing Hu, July, 2025.

AI is coming for your job.

At least, that's what a lot of big-shot CEOs and pundits keep insisting.

Ford's CEO, Jim Farley,

*Warns AI will wipe out half of white-collar jobs* – Fortune Article - July 5, 2025

Amazon's chief, Andy Jassy, has firmly stated the following in the Amazon official announcement:

*As we roll out more Generative AI and agents, it should change the way our work is done. We will need fewer people doing some of the jobs that are being done today* – Amazon Official Statement - June 17, 2025

And it's not just traditional execs?

Dario Amodei of AI startup Anthropic claimed

AI could wipe out half of all entry-level white-collar jobs — and spike unemployment to 10-20% in the next one to five years, Amodei told us in an interview from his San Francisco office. – Axios Original Interview - May 28, 2025

Yeah, it makes for a great headline. You can almost picture empty office chairs everywhere.

But hold on a second! Are you taking this at face value? We need to ask ourselves how much of this is genuine and how much is just marketing rhetoric.

Knowing the underlying technology, I'm not buying it, and neither should you.

The bold talk of mass automation often serves quite a different agenda than you'd thought (hint: cost-cutting), and the tech just isn't living up to the hype yet.

Let's break down what's really going on in plain English. Shall we?

## Automation ≠ Autonomy

People hear "AI agent" and imagine a digital employee, a software program that can understand goals, carry out tasks, adapt on the fly, and maybe even chat with other agents to get work done.

You could replace a whole department with these things.

Current "agents" are just large language models (LLMs) like GPT-4, wrapped in code and prompts to make them perform tasks. Think of this like super-efficient interns who have broad and generic knowledge of everything, **suffer from short-term memory loss, and suck at taking feedback**, have trouble taking even step-by-step instructions.

Cool? Absolutely. Someone you want to hire?

Maybe not…

Let's break down some of the most significant limitations (list not exhaustive) holding AI agents back from taking over real jobs. The unsexy truths that get lost in the hype.

## Brittle and over-scripted.

Today's agents have minimal true agency.

Their "initiative" is largely an illusion; behind the scenes, they follow (or are trying to) tightly choreographed steps that a developer or prompt writer set up.

If you ask an agent to do Task X, it will do X, then stop. Ask for Y, and it does Y. But if halfway through X something unexpected happens, say a form has a new field, or an API call returns an error, the agent breaks down.

Because **it has zero understanding of the task**.

Change the environment slightly (e.g., update an interface or move a button), and the poor thing can't adapt on the fly.

AI agents today lack a genuine concept of overarching goals or the common-sense context that humans use.

They're essentially text prediction engines.

If you tell a human customer support rep, "I'm locked out of my account," the human implicitly knows the goal is to get you back in, maybe verify your identity, check related issues, etc.

An AI agent doesn't inherently grasp that broader intent; it will respond only to the exact phrasing and info it's given.

It might reset your password, but it won't address the reason you're locked out, which may be a system outage, so it doesn't matter how many times you change your password; you still can't log in.

Complex, multi-step objectives confuse these agents because they can't truly plan like a person.

## Short-term memory.

Ever worked with someone who forgets instructions 10 minutes after you gave them?

That's your typical AI agent in a nutshell.

These models operate within a fixed "context window" (some can handle a lot of text but are still finite). **They don't remember past conversations or retain new knowledge** once the task is done. If an interaction stretches on too long or gets complex, they start losing the plot like a goldfish that circles the bowl and immediately forgets where it started .

Every session begins fresh, with the AI having no clue what it did an hour ago unless you explicitly feed that history back in. It's like Memento for machines – not exactly ideal for jobs that require continuity or accumulated expertise.

## Unpredictable outputs (aka non-determinism).

Run the exact same AI agent with the exact same input twice, and you'll get two different results.

Sometimes it's minor phrasing differences; other times, one attempt works, and the next fails miserably. This is because the underlying model can produce variations in each run. Traditional software is deterministic – give it input X, and you reliably get output Y every time.

That consistency lets engineers test and debug code methodically.

Not so with AI.

With non-deterministic behavior, testing becomes a nightmare!

Two questions will help you understand why this is such a big deal.

- How do you trust a system that might give a different answer or take a different action each time?
- How do you debug an error that you can't reliably reproduce?

It's like trying to catch a glitch in a program that rewrites itself on the fly.

This randomness is part of what makes AI creative, but it's also a huge barrier to using it for mission-critical, repetitive work where precision matters.

## Can’t Learn from Feedback

Here’s another big gap: today’s AI agents are terrible at learning from feedback in real-time.

Tell a human, “Hey, you did this wrong; do it differently next time,” and odds are, they’ll adjust (or at least try).

With AI Agents, you’re yelling into the void.

You can point out the mistake, correct the output, or even give step-by-step guidance…

But the agent won’t actually remember your feedback for next time.

Unless you explicitly code that correction into a new prompt or manually update the underlying model (which, let’s be honest, almost nobody does at the user level), it will make the same mistake again and again.

There’s no running “lessons learned” file. Every chat is a blank slate.

This means LLM-based AI agents are forever stuck in Day One mode, ie, good at generating plausible-sounding outputs but hopeless at evolving based on your real-world needs or corrections.

If you’re imagining a future where you “train up” your personal assistant like a new hire, forget it. Unless the technology has a breakthrough in building a world model in AI, for now, you’re just hitting the reset button every time you start a new task.

## They still need “adult” supervision.

Despite the “autonomous” branding, AI agents today require a human in the loop at crucial points.

Whether it’s a software script an AI wrote (did it actually work?) or an email to a client (is the tone right, are the facts correct?), there's still no company that can confidently say they trust the agent to send it off without a human double-check.

That means the AI isn't truly automating the job; it's automating parts of the job and handing the rest back to us humans.

With all that's been said, these AI agents can be powerful productivity boosters in the best cases but a cash burner at worst. They're great at "input provider" tasks, ie, generating draft content, summarizing info, and giving recommendations.

However, it failed dramatically when it came to making judgment calls or handling novel situations. They often faceplant.

Let me provide you with a couple of concrete examples of what happens when you unleash AI agents on real-world tasks.

## AI Agents Flunk Real-world Tests

To separate hype from reality, researchers have started benchmarking these AI agents.

The results are expected.

### Vals Finance Agent test

In fact, another benchmark (the Vals Finance Agent test) looked at tasks expected of an entry-level financial analyst. You know, reading financial reports, doing basic analysis across a range of AI models.

**No model managed to exceed 50% accuracy.**

The best was just about 48% correct (basically a coin flip), even after incurring a hefty cost in API calls. These aren't systems you'd trust to handle the workload of even a junior employee unsupervised.

### The Agent Company by a team at Carnegie Mellon

They essentially created a fake software company, complete with internal websites, databases, and even simulated co-worker agents. Then, let AI agents loose to run it.

It's the ultimate "AI, you think you can do this job? Go for it" test.

The tasks included writing and deploying code, fetching information, collaborating with other agents (such as asking a colleague for help), and so on. This is exactly the kind of multi-step, interdependent work humans do in a normal office.

So, did the agents crush it?

The most capable AI agent in the experiment managed to **complete only 24% of its assigned projects** on its own. In nearly three-quarters of the cases, it either outright failed or got so stuck that a human had to intervene and save the day.

Imagine if you hired a new employee and they could only do a quarter of their work without constant hand-holding, you wouldn't move on after their probation, would you? The detailed findings on how AI agents stumbled over things any reasonably competent human could handle.

For example, an agent ran into a simple pop-up window on a website, the kind with an "X" you click to close it. The poor thing got stuck there, unable to figure out how to proceed. Any human intern would instinctively dismiss a random pop-up and get on with their work, but the AI had no intuition for it.

AI agents also showed some hilariously absurd behavior when confused.

At one point, not knowing how to reach a particular person it was supposed to contact, an agent literally renamed another user account to impersonate the target person as if that solved the task! (Yes, it effectively tried to cheat and then declared victory!)

What these benchmarks lay bare is a **huge performance gap between AI marketing claims and real-world capability**.

There are hundreds of other proofs I can give you, but I'm pretty sure you get the point now.

It's one thing to demo an agent doing a canned task perfectly once. It's another to trust it with the messy assortment of work that makes up a real job day after day. They lack common sense, fall for simple traps, and often require more babysitting than a toddler.

# The Jobpocalypse That Backfired

When did the reality on the ground start to match the boardroom bravado?

### Misaligned employment stats

The U.S. economy added jobs in recent months, rather than shedding them, with approximately 147,000 new jobs in June, which helped lower unemployment to 4.1%. As you can see in the graphic below, basically nothing changed after ChatGPT's launch in late 2022.

If an AI takeover of the labor market were truly imminent, we'd expect to see the opposite trend (mass layoffs with no hiring), but that's not happening yet.

What's more, the actual adoption of AI at work is pretty low. One survey found that 43% of Americans use AI less than once a week, and barely 27% use it daily.

Instead, most people aren't heavily relying on it, which tells you something.

The "AI everywhere" narrative is sprinting far ahead of reality.

Not convinced?

Here's more.

## I'm being paid to fix issues caused by AI

Instead, 2025 has seen a dramatic about-face.

Firms that rushed to swap staff for AI are now spending even more to clean up the messes. The grand plan to save money with AI is *backfiring* in some expensive ways.

One striking, most recent example is **Sarah Skidd**, a product marketing writer from Arizona, who was hired to *fix* AI-generated copy for a client, as reported by the BBC.

Skidd said in the interview:

*I'm being paid to fix issues caused by AI*

Skidd ended up rewriting the entire copy from scratch, a task that took 20 hours at her rate of $100/hour, costing the client about $2,000.

Had a human written it to begin with, the bill likely would have been much lower.

Far from losing work to AI, Skidd says she's getting *more* business fixing these errors. And she's not alone. An emerging cottage industry of writers, engineers, and consultants is now devoted to tidying up AI's mistakes.

## Duolingo's High-Profile U-Turn

This irony extends to big companies.

In April 2025, language-learning app Duolingo announced it would become "AI-first," even vowing to *"gradually stop using contractors to do work that AI can handle"...*

A severe backlash ensued, and then, in May, Luis von Ahn was forced to walk back his stance. He clarified: "*To be clear: I do not see AI as replacing what our employees do (**we are, in fact, continuing to hire at the same speed as before**).*"

## Klarna's Customer Service Reality Check

Or, the Swedish fintech **Klarna** provides another cautionary tale of AI overhype.

In 2024, Klarna proudly announced that its new AI chatbot could *handle the workload of 700 customer service agents*, addressing approximately 75% of all customer chats.

The CEO, Sebastian Siemiatkowski, even boasted that he believed AI could perform essentially *all* the jobs humans do.

Fast forward a year, and the results were grim.

Customer satisfaction **plummeted by 22%** after the AI-only support rollout.

Siemiatkowski admitted that focusing too much on cutting costs had led to "lower quality" service and glaring **"empathetic gaps" that no algorithm could fill**.

By May 2025, Klarna quietly acknowledged the failure of its AI-first support strategy and began **hiring human support agents** again to restore service quality.

The feared AI-driven job purge has simply not materialized, period.

More and more organizations are finding that hastily implemented AI solutions **fail to deliver** on their promises. A recent S&P Global survey found that the share of companies scrapping *most* of their AI initiatives jumped to **42% in 2025** (up from just 17% the year before).

Nearly **46% of AI projects never make it out of the proof-of-concept stage** into production.

## AI Hype as a Business Strategy

It should be pretty clear now that agents are far from ready to wholesale replace humans in most jobs.

So why do we keep hearing the opposite?

Why are CEOs and commentators still beating the "AI will take your job" drum?

Simple: **misjudgment and management theater.**

### Misjudgment that no CEO likes to admit

Maybe misjudgment is a charitable explanation; some folks genuinely believe the hype.

They see rapid AI advances (and, to be fair, AI has improved very quickly in certain areas), and they assume we're only a few tweaks away from sci-fi-level autonomous workers.

It's a classic pattern throughout the last hundred years. I wrote about last year: [I Studied 200 Years' Of Tech Cycles. This Is How They Relate To AI Hype.]

I don't blame them. If you're a CEO surrounded by yes-men, shareholders eager to see stock price shoot with AI, and futurist consultants, you might truly think your company can go "50% AI staff" in a year or two.

However, this is a strategic error stemming from a **lack of understanding of the technology's current limitations**. Unfortunately, these CEOs are in for a rude awakening when the tech doesn't deliver.

## Typical cost-saving management theatre

Which, frankly, I suspect is driving a lot of the rhetoric.

We've been through an era of over-hiring and cheap money, and now reality is biting. But nobody likes to be the bad guy who lays off thousands of workers just to bump the stock price.

So, AI narrative provides a handy cover story.

By claiming, "We're not downsizing, we're modernizing," companies can soften the PR blow.

It's a way of saying, "It's not that we want to cut jobs, it's that this exciting new tech just makes it inevitable!" This narrative can also goose investor excitement. Announcing a big AI initiative with a round of cost cuts as the cherry on top, investors expect their stock to tick up as Wall Street salivates over efficiency gains.

It's corporate spin 101.

AI, in 2025, is still the ultimate buzzword to paper over the boosting of the bottom line's ugly motive.

In reality, when we peel back the spin, we often find that nothing replaced those workers at all; the remaining humans just have to pick up the slack, or some other tasks simply get dropped.

In cases where AI is introduced, it typically assists with tasks (such as helping employees draft an email more efficiently), something that barely tickles the proactivity itch.

Not to mention, most of these firms don't even have the proper infrastructure, data integration, and customization required, even when an advanced AI model arrives.

## You Won't Be Replaced by AI But by Hype.

At the end of the day, AI isn't necessarily taking your job, but the hype is.

Don't get me wrong, it can be helpful, particularly in specific use cases.

But the notion that we're on the brink of a wholesale AI takeover is bullshit that media and AI companies CEO try to force down our throats.

AI agents today are amazing tools for mini-automation (simple tasks that don't take up to 15 mins), not wholesale job replacement. They can help with some steps in your workflow, that's about it.

But they are nowhere near ready to be employees on their own.

Even the smartest AI agents can do is still vast, especially when it comes to judgment, adaptability, and responsibility.

Of course, this doesn't mean AI won't keep getting better or that it won't eliminate some roles.

But as someone who was in the trenches, connecting software with AI models, I can tell you we're not there today or tomorrow, or next financial year.

If you're still so worried about being replaced by AI in the coming years, here’s some advice I have concluded based on the latest research after reading hundreds of papers and interviewing dozens of researchers.

If you have more responsibility in the left column in the sheet (link provided) below, try to move your day-to-day duties to the right column slowly.

When it comes to the daily grind of real jobs, the robots still have a lot to learn.

## Editor’s Note

Jing Hu is from Taiwan and is currently based in London, UK. She prides herself on deep articles that are accessible, stay beyond the hype and are fluff and jargon-free. I trust her research and enjoy her critical takes on topics. Maybe you would too.